



Forces and Newton's Laws of Motion

• Worksheet 1

Mechanics Chapter 3: Forces and Newton's Laws of Motion. Answer all questions. Show your working clearly. Total: 83 marks. Time: 105 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are four sections, one for each topic in Chapter 3. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Take $g = 9.8 \text{ m s}^{-2}$. Draw a clear force diagram for every question, and give answers to 3 significant figures.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Types of force and force diagrams	17	
B Newton's first and second laws	20	
C Newton's third law and connected particles	20	
D Pulleys	21	
Roar-level challenge	5	
Total	83	

Section A Types of force and force diagrams

17 marks

1. A book rests on a horizontal table. Draw a force diagram showing the forces acting on the book, and name each force. [2]



2. A box is pulled along rough horizontal ground by a horizontal rope. Draw a force diagram showing all four forces acting on the box. [2]

3. A particle of mass 3 kg rests on a horizontal floor. [3]

(a) Find the normal reaction on the particle.

(b) A vertical force of 10 N is now applied to the particle, acting upwards. Find the new normal reaction.

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4. Three horizontal forces act on a particle: 5 N due east, 8 N due west and 12 N due north. [3]
Find the magnitude of the resultant force and its bearing.

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5. A particle of mass 4 kg rests in equilibrium on rough horizontal ground. It is pulled by a light string with tension 20 N, inclined at 30° above the horizontal.

(a) Draw a force diagram showing all the forces on the particle. [2]

(b) Find the normal reaction between the particle and the ground. [2]

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(c) Find the magnitude of the frictional force. [2]

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(d) Explain why the normal reaction is not equal to the weight of the particle. [1]

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(Total for Question 5 is 7 marks)



Section B Newton's first and second laws

20 marks

6. A resultant force of 12 N acts on a particle of mass 3 kg. Find its acceleration. [2]

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7. A box of mass 5 kg is pulled along horizontal ground by a horizontal force of 30 N against a resistance of 10 N. The box starts from rest. [3]
Find the acceleration of the box and its speed after 3 s.

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8. A car of mass 1200 kg has a driving force of 3000 N and experiences a resistance of 600 N. [3]
Find the acceleration of the car, and the time it takes to reach 20 ms^{-1} from rest.

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9. A particle of mass 2 kg lies on a smooth horizontal surface. It is pulled by a force of 20 N acting at 30° above the horizontal. [4]
Find the acceleration of the particle and the normal reaction from the surface.

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10. A lift of mass 500 kg carries a man of mass 80 kg. The lift is raised by a vertical cable.

(a) The lift accelerates upwards at 1.2 m s^{-2} . Find the tension in the cable. [3]

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(b) Find the reaction of the floor of the lift on the man. [2]

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(c) Later, the lift is moving upwards and decelerating at 2 m s^{-2} . Find the reaction of the floor on the man. [2]

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(d) Explain what Newton's first law tells you about the forces when the lift moves at constant speed. [1]

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(Total for Question 10 is 8 marks)



Section C Newton's third law and connected particles

20 marks

11. A book rests on a table. Explain why the weight of the book and the normal reaction on the book are **not** a Newton's third law pair. [2]

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12. A car of mass 1000 kg tows a trailer of mass 400 kg using a light rigid towbar along a horizontal road. [3]
The driving force is 2100 N and there is no resistance to motion.
Find the acceleration and the tension in the towbar.

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13. Particles A (3 kg) and B (2 kg) lie on a smooth horizontal table, joined by a light inextensible string. [3]
A horizontal force of 15 N is applied to A , directly away from B .
Find the acceleration of the particles and the tension in the string.

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- 14.** A car of mass 1200 kg tows a trailer of mass 300 kg along a horizontal road using a light rigid towbar. **[4]**
The resistances to motion are 400 N on the car and 200 N on the trailer. The driving force is 3600 N.
Find the acceleration and the tension in the towbar.

- 15.** A truck of mass 2000 kg tows a trailer of mass 1000 kg along a straight horizontal road using a light rigid towbar. The resistances to motion are 1000 N on the truck and 500 N on the trailer.

- (a)** The driving force of the truck is 6000 N. Find the acceleration and the tension in the towbar. **[4]**

- (b)** The driving force is removed and the truck's brakes are applied, giving an extra braking force of 2500 N on the truck. The resistances are unchanged. Find the deceleration, and the force in the towbar, stating whether it is a tension or a thrust. **[4]**

(Total for Question 15 is 8 marks)



Section D Pulleys

21 marks

16. Particles of mass 5 kg and 3 kg are joined by a light inextensible string passing over a smooth fixed pulley. They hang freely and are released from rest. Find their acceleration. [2]

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17. Particles of mass 6 kg and 4 kg hang freely either side of a smooth fixed pulley, joined by a light inextensible string. They are released from rest. Find the acceleration and the tension in the string. [3]

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18. A particle of mass 4 kg lies on a smooth horizontal table. It is joined by a light inextensible string, passing over a smooth pulley at the edge of the table, to a particle of mass 1 kg hanging freely. Find the acceleration and the tension in the string. [3]

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19. Particles of mass 5 kg and 3 kg hang either side of a smooth pulley and are released from rest. The 5 kg particle is 1.5 m above the floor. Find the speed with which it hits the floor, and the time taken. [4]

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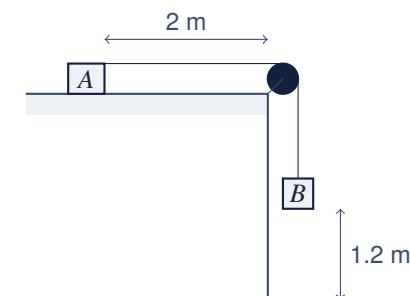
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20. Particle A of mass 2 kg lies on a smooth horizontal table, 2 m from a smooth pulley P at the edge. It is joined by a light inextensible string passing over P to a particle B of mass 3 kg , which hangs 1.2 m above the floor. The system is released from rest.



- (a) Find the acceleration of the system and the tension in the string. [4]

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- (b) Find the speed of B as it hits the floor. [2]

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- (c) B does not bounce. Find the total time from release until A reaches the pulley. [3]

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(Total for Question 20 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

21. Particles of mass 4 kg and m kg, where $m < 4$, hang freely either side of a smooth fixed pulley, joined by a light inextensible string. When released from rest, they move with acceleration 2.8 m s^{-2} . Find m and the tension in the string. [5]

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END OF WORKSHEET

Total: 83 marks